

Assignment-1

a) Complete the table with

$$\begin{array}{r|l} 2 & 143.75 \\ \hline & 71 \end{array}$$

1) 143.75

$$\begin{array}{r|l} 2 & 143 & 1 \\ \hline 2 & 71 & 1 \\ \hline 2 & 35 & 1 \\ \hline 2 & 17 & 1 \\ \hline 2 & 8 & 0 \\ \hline 2 & 4 & 0 \\ \hline 2 & 2 & 0 \\ \hline & 1 & \end{array}$$

$$\begin{array}{r} 0.75 \times 2 \\ \hline 1.50 \\ 0.50 \times 2 \\ \hline 1.00 \end{array}$$

$$143.75 = (10001111.11)_2$$

b) octal conversion

$$\begin{array}{r|l} 8 & 143 & 7 \\ \hline 8 & 17 & 1 \\ \hline & 2 & \end{array}$$

$$\begin{array}{r} 0.75 \times 8 \\ \hline 6.00 \end{array}$$

$$= (217.6)_8$$

3) Hexa conversion

$$\begin{array}{r|l} 16 & 143.75 \\ \hline & 00 \end{array}$$

$$\Rightarrow 143.75$$

$$\begin{array}{r|l} 16 & 143 & 15 \\ \hline 16 & 8 & \end{array}$$

$$\begin{array}{r} 0.75 \times 16 \\ \hline 12.00 \\ 0.75 \times 16 \\ \hline 12.00 \end{array}$$

$$\begin{array}{r} 0.75 \times 16 \\ \hline 12.00 \\ 0.75 \times 16 \\ \hline 12.00 \end{array}$$

$$\Rightarrow (8F.C)_{16}$$

2) $(11011.101)_2 \Rightarrow$ Decimal, Octal, Hexa

Decimal

$$1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 + 0 \times 0.5 + 1 \times 0.25$$

$$(27.25)_{10}$$

Octal

$$011011.101$$

$$\Rightarrow (33.5)_8$$

Hexa

$$00011011.1010$$

$$\rightarrow (1B.A)_{16}$$

$$\begin{array}{r} 64 \times 4 \\ 256 \end{array}$$

3) $(453)_8 \Rightarrow$ Decimal, Binary, Hexa

$$4 \times 64 + 5 \times 8 + 3 = (299)_{10}$$

$$(453)_8 = (100101011)_2$$

$$000100101011 \Rightarrow (12B)_{16}$$

4) IF.C = decimal, Binary, Octa

$$1 \times 16 + 15 \times \frac{18}{16} \times 3$$

$$18 \times 4$$

$$(31.75)_{10}$$

$$\begin{array}{r} 0.75 \\ 1) 30 \\ 28 \\ \hline 20 \end{array}$$

$$(00011111.1100)_2$$

$$00011111.1100$$

$$37.60$$

$$\Rightarrow (37.60)_8$$

baud 10
1 Sec 1000 bit

Q. Find equivalents

a) $(324)_5$

$25 \times 8 + 10 + 4 = 75 + 10 + 4 = 89$

b) $(32.23)_4 =$

$12 + 2.0.5 + 0.75$

14.

15.25

c) $(231)_{10} = ?_7$

7	231	0
7	33	5
	4	

$= (450)_7$

d) $(189)_{10}$

189	3
31	1
5	

$= (513)_6$

Q. Find the radix:

$(112120)_r = (131)$

Q. Addition

a) $1101_2 + 11011111_2$

~~11011111~~
 ~~1101~~
 1100100

$\Rightarrow$

11011111
 1101
 11101100

b) $77652 + 96543$

$$\begin{array}{r} 77652 \\ 96543 \\ \hline 176415 \end{array}$$

c) $578A$
ABDE
10368

d) 234
 315
553

Q5 Binary $\rightarrow$ Gray Code

a) 10101
 $=11111$

b) 01110
 $=01001$

Q6 Gray code $\Rightarrow$ Binary

a) $01101 \Rightarrow 01001$

b) $10101 \Rightarrow 11001$

Q. Find the radix of

$$a) (312/80)_n = (13.1)_n$$

$$(312 + r + 2) = (310)_n$$

$$(80)_n = 2n$$

$$(13.1)_n = n + 3 + \frac{1}{n}$$

$$\frac{312 + r + 2}{2n} = n + 3 + \frac{1}{n}$$

$$312 + r + 2 = 2n^2 + 6n + 2$$

$$3n^2 - 2n^2 + 6n - 6n + 2 - 2 = 0$$

$$n^2 - 5n + 10 = 0$$

$$n^2 - 5n$$

$$n^2 - 5n = 0$$

$$n(n - 5) = 0$$

$$\underline{n = 5} \quad \underline{\text{radix} = 5}$$

$$b) 264_n + 136_n = 433_n$$

$$(264)_n = 2n^2 + 6n + 4$$

$$(136)_n = 1n^2 + 3n + 6$$

$$(433)_n = 4n^2 + 3n + 3$$

$$2n^2 + 2n + 4 + n^2 + 3n + 6 = 4n^2 + 3n + 3$$

$$3n^2 + 4n + 10 = 4n^2 + 3n + 3$$

$$n^2 - 6n - 7 = 0$$

$$n^2 + n - 6n - 7 =$$

$$n^2 + n - 7n - 7 =$$

$$n(n+1) - 7(n+1)$$

$$n = -1 \quad \underline{n = 7}$$

c) $221505_8 = (12345)_r$

$$12345 = n^4 + 2n^3 + 3n^2 + 4n + 5$$

$$221505 =$$

$$256 \times 2 + 64 \times 5 + 16 \times 1 + 4 \times 1$$

$$256 \times 2 + 512 \times 2 = 0$$

$$(221505)_8 = 74565$$

$$n^4 + 2n^3 + 3n^2 + 4n + 5 = 74565$$

$$n = 16$$

$$(16)^4 + 2(16)^3 + 3(16)^2 + 4(16) + 5 = 74565$$

$$65536 + 8192 + 768 + 64 + 5 = 74565$$

then

$$\underline{\underline{n = 16}}$$

$$\frac{-7n \pm \sqrt{49n^2 - 4(1)n}}{2n}$$

$$\frac{-7n \pm \sqrt{49n^2 - 4n}}{2n}$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{-1 \pm \sqrt{1 - 4 \cdot 2 \cdot 8}}{2}$$

$$\frac{-1 \pm \sqrt{1 - 64}}{2}$$

$$\frac{+6 \pm \sqrt{36 + 28}}{2}$$

$$\frac{3 \pm \sqrt{64}}{2}$$